

age	income	student	credit-rating	buys-computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$\bullet \text{ Info}(D) = - \sum_{i=1}^m P_i \log_2(P_i)$$

$$\bullet \text{ Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$\bullet \text{ Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

Age	Class		
	yes(P)	no(N)	(P,N)
<=30	2	3	0.971
31...40	4	0	0
>40	3	2	0.971

$$L = 30 - 5$$

$$\text{Yes} = 2; \text{No} = 3$$

$$\Rightarrow 31 \dots 40 - 4$$

$$\text{Yes} = 4; \text{No} = 0$$

$$\Leftrightarrow > 40 - 5$$

$$\text{Yes} = 3; \text{No} = 2$$

$$1) I(P, N) = I(2, 3)$$

$$= -\frac{2}{5} \log_2(2/5) - \frac{3}{5} \log_2(3/5) = 0.971$$

$$2) I(4, 0) \Rightarrow I(P, N)$$

$$= \frac{4}{4} \log_2(4/4) - \frac{0}{4} \log_2(0/4) = 0$$

$$3) I(P, N) = I(3, 2)$$

$$= -\frac{3}{5} \log_2(3/5) - \frac{2}{5} \log_2(2/5)$$

Total:

$$I(P, N) = I(9, 5)$$

$$= -\frac{9}{14} \log_2(9/14) - \frac{5}{14} \log_2(5/14) = 0.94011$$

$$\cdot \text{Info}(D) = -\sum_{j=1}^7 \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$= \frac{15}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971$$

$$= 0.6934 \approx 0.694$$

$$\text{Gain}_{\text{age}} = 0.940 - 0.694$$

$$= 0.246 //$$

Gain for Income:

Income	P	N		
high	2	2	[1]	- 4
medium	4	2	[0.918]	- 6
Low	3	1	[0.811]	- 4
	9	5	<u>0.940</u>	(0.029)

$$1) I(P, N) = I(2, 2) = -\frac{2}{4} \log(2/4) - \frac{2}{4} \log(2/4) = 1.$$

$$2) I(P, N) = I(4, 2) = -\frac{4}{6} \log(4/6) - \frac{2}{6} \log(2/6) = 0.918 //$$

$$3) I(P, N) = I(3, 1) = -\frac{3}{4} \log(3/4) - \frac{1}{4} \log(1/4) = 0.811.$$

Total

$$I(P, N) = I(9, 5) = -\frac{9}{14} \log(9/14) - \frac{5}{14} \log(5/14) = 0.940 //$$

$$\underline{\text{Info}(D)} = \sum_{j=1}^4 \frac{|D_j|}{|D|} \times \text{Info } |D_j|$$

$$= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811 = 0.9108$$

$$\text{Gain Income} = 0.940 - 0.9108 = 0.0292 //$$

Info Gain

age 0.246

Income 0.029

Student 0.151

credit 0.048

$$\cdot \text{Split Info} = - \sum_{j=1}^j \frac{|D_j|}{|D|} \times \log_2 \times \frac{|D_j|}{|D|}$$

For the attribute age:

$$\text{Gain Ratio of age} = \frac{\text{Gain(age)}}{\text{Split Info(age)}}$$

$$\begin{aligned} \text{Split I} &= - \left[\frac{5}{14} \log_2 \left(\frac{5}{14} \right) + \frac{4}{14} \log_2 \left(\frac{4}{14} \right) + \frac{5}{14} \log_2 \left(\frac{5}{14} \right) \right] \\ &= 1.5774 \end{aligned}$$

$$\text{Gain Ratio} = \frac{0.246}{1.5774} = 0.156 //$$

Income:

$$gini(D) = 1 - \sum_{j=1}^j (P_j^2)$$

$$gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$$

Gini Index of Income = $1 - \sum (P_j^2)$

$$= 1 - [$$

$$[gini = 0.459]$$

$$\begin{aligned} gini_A^{LM}(D) &= \frac{10}{14} \left[1 - \left[\left(\frac{7}{10} \right)^2 + \left(\frac{3}{10} \right)^2 \right] \right] + \frac{4}{14} \left[1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] \right] \\ &= \frac{10}{14} \times 0.42 + \frac{4}{14} \times 0.50 \\ &\approx 0.4429 // \end{aligned}$$

$$D_1 = LM; D_2 = L$$

$$\begin{aligned} Gini(D) &= \frac{10}{14} \left[1 - \left(\left(\frac{6}{10} \right)^2 + \left(\frac{4}{10} \right)^2 \right) \right] + \left(\frac{4}{14} \right) \left[1 - \left(\left(\frac{6}{10} \right)^2 + \left(\frac{4}{10} \right)^2 \right) \right] \\ &= \frac{10}{14} \times 0.48 + \frac{4}{14} \times 0.375 \\ &= 0.4500 // \end{aligned}$$

$$D_1 = LH, D_2 = M$$

$$\begin{aligned} Gini(D) &= \frac{8}{14} \left[1 - \left(\left(\frac{5}{8} \right)^2 + \left(\frac{3}{8} \right)^2 \right) \right] + \frac{6}{14} \left[1 - \left(\left(\frac{4}{6} \right)^2 + \left(\frac{2}{6} \right)^2 \right) \right] \\ &= \frac{8}{14} [0.4688] + \frac{6}{14} [0.444] \\ &= 0.4583 // \end{aligned}$$